

# LeLunar: Learning the Moon by Predicting One Instrument from Another

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## Abstract

Orbital missions have mapped the Moon with many instruments, from multispectral images taken under changing sunlight to static maps of topography, roughness and composition. The products are co-registered but differ in resolution, units and physics, and labels are scarce. We present LeLunar, a self-supervised model that learns from this stack by predicting, in representation space, what one instrument would record at a 60 km footprint given another instrument's view of it. Per-product stems give every product the same ground cell per token, from 60 m to 1 km per pixel, and a cross-attention predictor, told the target product and its sun geometry, predicts the token grid of a moving-average encoder. Training this way has a failure mode that the loss hides. Several runs converged to a position code, in which tokens depend on their place in the grid and hardly on the terrain, while the loss fell tenfold and within-image rank and VICReg penalties improved. We explain why these signals are blind, show that a covariance penalty and a variance hinge on predictions push toward the shortcut, and propose two cross-sample monitors that flagged all seven failures we analyzed. Frozen LeLunar features detect craters on par with a masked-token lunar foundation model and gain up to 0.23 mAP<sub>50</sub> from a second input product. Probes recover relief and albedo from single images, and predicted shadows follow the sun the predictor is given (AUC 0.89 with the true sun, 0.59 with another).

## 1. Introduction

Few places beyond Earth have been mapped as thoroughly as the Moon. The Lunar Reconnaissance Orbiter has returned camera images [71], laser altimetry [77], thermal radiometry [59] and radar [56] since 2009. Kaguya contributed multiband imagery and stereo topography [46, 57], and GRAIL a gravity field [98]. For almost any patch of ground one can look up a stack of co-registered maps. They record its brightness at five wavelengths under a given sun, its height and slope, its roughness over tens of meters and its titanium or olivine content. What is scarce, as elsewhere in

remote sensing, is labels.

Earth observation has answered that scarcity with foundation models pretrained on unlabeled multi-sensor archives [6, 23, 31, 44, 85, 91]. The Moon got its first ones in 2026 [30, 33, 64], and all three learn by reconstructing masked pixels or discrete tokens. We ask what a model learns if it must instead predict, in representation space, what one instrument would record at a place, given what another instrument saw there.

Lunar data suit the question for two reasons. The products are physically coupled. There is no atmosphere, so the shading in an image is set mostly by topography and the position of the sun, and albedo and roughness explain much of the rest. And every image comes with its illumination geometry, while the same terrain has been imaged many times under different suns. A predictor that is told the sun can therefore be asked what a footprint looks like under a particular light. To answer, the representation of the topography must carry whatever casts the shadows.

We build this as a joint-embedding predictive architecture (JEPA) [3, 13]. LeLunar encodes one product of a 60 km footprint with part of it hidden. A predictor, told the target product and the target's sun, then predicts the token grid that a slowly updated copy of the encoder produces for another product of the same footprint (Fig. 1). Each product, from 60 m to 1 km per pixel, gets a stem that reads its native pixels. As in OmniSat and AnySat [5, 6], a token then covers the same ground cell whatever its product, but we keep one token grid per product.

Most of our effort went into getting this to train. Cross-modal latent prediction has an escape route that masked prediction within one image mostly lacks. Much of the target grid cannot be predicted from the source at the scale of a token, and the student and its target share cheap channels: token position, product identity and footprint location. In several runs the model settled into a state in which every token encoded little more than its place in the grid. The loss fell tenfold, the effective rank within a tile rose and the VICReg penalties improved, while the representation stopped telling one footprint from another. This position-code collapse is a cousin of the positional collapse CAPI [26] saw in

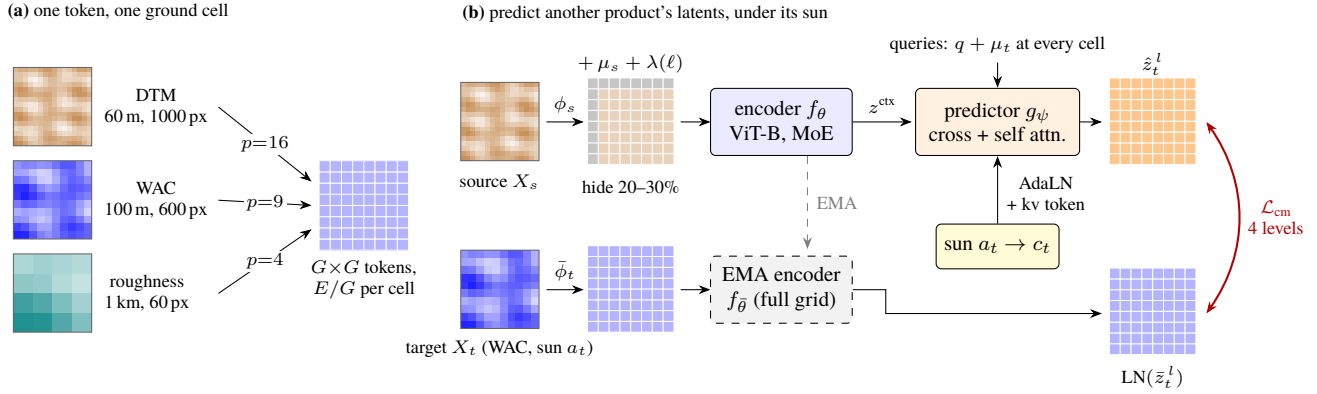


Figure 1. LeLunar in one picture. **(a)** Each product has its own stem, whose kernel equals its native pixels per token, so a 60 m DTM, a 100 m WAC image and a 1 km roughness map all become the same  $G \times G$  grid of ground cells. **(b)** The encoder sees the source with 20–30% of its tokens hidden. The predictor’s queries carry the target product, and the target’s sun conditions every predictor layer through AdaLN and an extra key/value token. The target is the EMA encoder’s normalized output for the other product at four depths. The dense loss on visible source tokens (Eq. (5)) is not drawn.

latent masked image modeling, here with continuous targets and across modalities.

Our contributions are:

- A cross-modal JEPA for co-registered planetary rasters, in which a predictor told the target product and the sun of the target observation predicts the dense token grid of another product (Sec. 3).
- An account of collapse in this setting from several dozen pretraining runs. It shows why loss, within-sample rank and VICReg miss a position code, and that the VICReg covariance penalty and a variance hinge on predictions push toward it (Secs. 4 and 5.1).
- Two cheap cross-sample monitors, which flagged all seven failures we analyzed, together with their blind spots (Sec. 4.3).
- Frozen features that detect craters on par with a masked-token lunar foundation model and gain up to 0.23 mAP<sub>50</sub> from a second input product. Relief, albedo and shadow probes show that the predictor uses the sun it is given (Sec. 5.4).

## 2. Related work

**Latent prediction.** I-JEPA [3] predicts the representations of masked image blocks from a visible block, with targets from an exponential moving average (EMA) of the encoder, a teacher also used by BYOL, DINO, iBOT and data2vec [9, 19, 34, 58, 96]. V-JEPA and V-JEPA 2 do the same for video [4, 13], and V-JEPA 2.1 [55] adds supervision at four depths and a distance-weighted loss on visible tokens, both of which we adopt. Multimodal masked modeling [8, 53] predicts across modalities but reconstructs pixels or tokens. Predicting one modality’s latents from another’s dates back at least to audio-visual RAVen [35]. In Earth ob-

servation the closest mechanism is Galileo’s [85]. Its predictor queries carry the target channel group and cross-attend to the other groups, with EMA targets, but inputs are re-sampled to 10 m and the loss is contrastive. AnySat [6] predicts fused multi-sensor cell embeddings. X-JEPA, CR-JEPA and Le MuMo JEPA [21, 24, 39] predict or regularize pooled embeddings, mostly for retrieval. To our knowledge no JEPA conditions its predictor on the illumination geometry of the target. The nearest precedents condition on the month (Galileo), the target view’s projection [63] or a caption [41].

**Collapse and its diagnosis.** Joint-embedding methods can collapse completely or into a low-dimensional subspace [40, 45, 83]. VICReg [12] and Barlow Twins [95] decorrelate features across a batch. C-JEPA [54] adds VICReg to I-JEPA. LeJEPA [10] replaces the teacher with a sliced test toward an isotropic Gaussian (SIGReg), whose gradients can vanish under collapse [89]. RankMe [32] and LiDAR [82] estimate quality from the embedding spectrum, and Latent MIM [88] monitors the cosine between pooled embeddings of different images. Objectives can also be met without content. Context prediction learned patch location from chromatic aberration [27], and JEPAs trained with VICReg can latch onto slow, trivially predictable features [78]. CAPI [26] reports positional collapse, in which latent masked-image models predict the position of masked tokens instead of their content. It was found by inspecting feature maps and fixed by clustering targets per position. StoP [11] and SiamJEPA [92] make positions stochastic or shuffle the teacher. OlmoEarth [38] replaced EMA targets, which collapsed in its experiments, with a frozen random projection. Absolute position also leaks through

zero padding [22, 43, 47], and self-supervised ViTs, including ones with rotary embeddings, carry input-independent positional components [61, 94]. We characterize positional collapse in a continuous-target, cross-modal JEPA and show which monitors and regularizers miss it or reward it.

**Resolution in Earth observation models.** Most multi-sensor models resample their inputs to a common grid [31, 44, 85, 91]. The alternatives are GSD-scaled positional codes [29, 42, 68] and per-band or per-sensor patch sizes [42, 87]. OmniSat and AnySat [5, 6] cut all modalities into the same ground cells, and we follow them. They fuse a cell’s modalities into one token, however, which rules out predicting one product from another. Location encoders on the sphere [48, 52, 72, 86] provide a geographic prior. We use Slepian functions [76], as proposed for location encoding by Rao *et al.* [67].

**The Moon and other planets.** Lunar machine learning has mostly meant task-specific models. Examples are crater detection [75, 81, 93], rockfall and boulder mapping [7, 15, 16], imaging of permanently shadowed regions [17] and single-view topography [20, 73]. The NASA-IBM lunar foundation model [30] retrains TerraMind’s masked-token recipe on WAC and NAC imagery and topography from SomBench [60]. It takes acquisition geometry as encoder input, and static maps enter it as tile-averaged scalars. LunarFM [33] is a MultiMAE over 18 global channels on 15 km chips and names JEPA-style objectives as future work. Moonstone [64] is a modality-grouped MAE at 237 m. On Mars, MOMO [66] merges per-sensor MAEs, and Mars-Bench [65] standardizes evaluation. None of these models predicts latents across modalities. LeLunar keeps static maps as spatial inputs and targets at native resolution, and learns from repeat observations under different suns.

### 3. Data and model

#### 3.1. Co-registered lunar footprints

Our data are 60 km  $\times$  60 km footprints of the lunar surface, with every available orbital product cut to the same ground extent and map projection. A footprint appears once per Wide Angle Camera (WAC) observation, so its static maps are paired with images taken under different suns. The full set has 10,629 footprints and 476,635 rows. Table 1 lists the products, from 60 m per pixel (topography, Kaguya Multi-band Imager) to 1 km (LOLA roughness). Diviner bolometric temperature and GRAIL gravity are too coarse to tokenize and serve only as an auxiliary target (Sec. 3.6). Each WAC row stores the incidence, emission and phase angles and the sub-solar azimuth, latitude and longitude of the acquisition. Splits follow Lunar Transverse Mercator zones,

Table 1. Products used in this paper.  $r$ : native ground sample distance.  $C$ : bands.  $p$ : native pixels per token at  $G=64$  and  $S_{\max}=1000$ , before and after a kernel floor of 4 (Sec. 3.3). \*The appearance depends on the sun, and each observation carries its illumination angles. The lower block joins in the 12-product runs.

| Product                | Source              | $r$ (m) | $C$  | $p$               |
|------------------------|---------------------|---------|------|-------------------|
| WAC visible*           | LROC WAC [71]       | 100     | 5    | 9                 |
| WAC ultraviolet*       | LROC WAC            | 500     | 2    | 2 $\rightarrow$ 4 |
| Elevation              | SLDEM2015 [14]      | 60      | 1    | 16                |
| Slope; aspect          | from elevation      | 60      | 1; 2 | 16                |
| 643 nm mosaic          | WAC normalized [18] | 100     | 1    | 9                 |
| Roughness              | LOLA [49]           | 1000    | 1    | 1 $\rightarrow$ 4 |
| Morphologic mosaic     | WAC [79]            | 100     | 1    | 9                 |
| Normalized reflectance | WAC normalized [18] | 500     | 7    | 2 $\rightarrow$ 4 |
| MI reflectance         | SELENE MI [50]      | 60      | 8    | 16                |
| MI mineralogy          | SELENE MI [50]      | 60      | 7    | 16                |
| TiO <sub>2</sub>       | WAC [74]            | 400     | 1    | 2 $\rightarrow$ 4 |

so that no validation footprint overlaps a training one. We exclude polar footprints. **[TODO: non-polar split counts]**

Values outside gross physical bounds are marked missing, the rest are clipped to ranges set by instrument scientists, and heavy-tailed products are log-transformed. Rasters larger than  $S_{\max}$  pixels per side are area-averaged down, ignoring missing pixels. Every band is standardized with training statistics, and missing pixels are then set to zero, the mean.

#### 3.2. Task

For each row we shuffle the available products and take the first as the source  $s$  and the next  $K=2$  as targets  $t_1, t_2$ . The source is encoded with part of its token grid hidden. For each target the model predicts the full token grid that an EMA of the encoder produces from the target raster, which the student never sees. It is told only which product to predict and, for WAC targets, the sun of that observation. Since each footprint has several WAC rows, the same topography must be turned into images under different suns.

#### 3.3. One token, one ground cell

We started by resampling all products to a shared  $4G \times 4G$  canvas with  $4 \times 4$  patches. At  $G=64$  this shrinks the 1000 px topography raster four times and the 600 px WAC image 2.3 times, and blows the 60 px roughness map up four times. We now align tokens to ground cells, as OmniSat and AnySat do [5, 6], and give every product  $m$  its own stem  $\phi_m$ , whose kernel and stride equal its native pixels per token,

$$p_m = \max(k_{\min}, \text{round}(\min(E/r_m, S_{\max})/G)), \quad (1)$$

where  $E=60$  km is the footprint extent. The raster is resized to  $Gp_m$  pixels per side, which changes it by a few percent unless the floor below applies, and  $\phi_m$  maps it to a  $G \times G$  grid of  $D$ -dimensional tokens. Every token is then one

ground cell of  $E/G$  meters (940 m at  $G=64$ ), whatever the product.

**Kernel floor.** At  $G=64$ , Eq. (1) with  $k_{\min}=1$  gives LOLA roughness one pixel per token. Each token is then a scalar times a fixed vector, so all roughness tokens lie on a line in  $\mathbb{R}^D$  (measured effective rank 1), a target that is easy to fit and carries little. A floor  $k_{\min}=4$  pre-upsamples coarse products so that each token sees a  $4 \times 4$  patch.

**Convolutional stems.** A linear patch projection [28] must preserve edges and texture through one matrix, so our latest recipe uses factorized convolutional stems [90].  $p_m$  is factored into stride-3 and stride-2 stages ( $9=3 \cdot 3$ ,  $16=2^4$ ,  $4=2^2$ ), each a convolution followed by GroupNorm and GELU, with channel width doubling toward  $D/2$ , then a  $1 \times 1$  projection to  $D$ . The usual  $3 \times 3$  kernel with padding 1 has a quiet flaw here, a case of the receptive-field offsets that stride and padding produce [1, 2]. It centers output  $o$  of a stride- $s$  stage on input pixel  $so$  rather than on the center  $so + (s-1)/2$  of its cell. The shifts add up across stages to

$$\delta_m = -\frac{p_m - 1}{2p_m} \text{ cells}, \quad (2)$$

which is  $-0.38$ ,  $-0.44$  and  $-0.47$  of a token for  $p_m = 4, 9, 16$ . Because the shift depends on  $p_m$ , tokens of different products refer to ground up to 0.09 cells apart. A kernel of size  $s+2$  with padding 1 centers the output on its cell, overlaps its neighbours at every stride and keeps the exact  $G \times G$  output. We use it in all stride-2 and stride-3 stages (Appendix A).

### 3.4. Encoder, location and experts

The encoder  $f_\theta$  is a ViT-B [28] (12 blocks,  $D=768$ , 12 heads). Positions enter only through axial 2D rotary embeddings [37, 80] on queries and keys. The last four blocks replace the MLP with a sparse mixture of experts: 8 experts, top-2 routing, a router with a learned bias per product, and the V-MoE load-balancing loss [69]. A token enters the encoder as

$$e_{m,i} = \text{LN}(\phi_m(X_m)_i) + \mu_m + \lambda(\ell), \quad (3)$$

where  $\mu_m$  is a learned product embedding and  $\lambda(\ell)$  encodes the footprint center  $\ell$  with 1536 Slepian functions [67, 76] (band limit 140) and an MLP. Only the tokens of the visible set  $V$  reach the encoder.  $V$  is one square block covering 70–80% of the grid, drawn per batch, so 20–30% of the source is hidden. The encoder returns layer-normalized features after blocks 3, 6, 9 and 12. A linear layer fuses them into the context  $z^{\text{ctx}} \in \mathbb{R}^{|V| \times D}$ .

The target branch is an EMA copy of the whole input pipeline: stems, product and location embeddings, and en-

coder. It reads the full, unmasked target raster and returns four layer-normalized feature grids  $\bar{z}_t^l \in \mathbb{R}^{G^2 \times D}$ .

### 3.5. A predictor that is told the sun

The predictor  $g_\psi$  is a 12-layer decoder as wide as the encoder. I-JEPA keeps its predictor narrow [3], but translating between products is harder than filling in part of one image. Its  $G^2$  queries are a learned vector plus the target product embedding, made distinct by rotary embeddings of their cells. Each layer cross-attends from the queries to the context, self-attends among the queries and applies an MLP. The illumination angles  $a_t$  are encoded as cosines and sines of incidence, emission, phase, sub-solar azimuth and longitude, plus sub-solar latitude and a validity bit, which is off for static targets. A small MLP turns this into a condition vector  $c_t$  that sets the scale and shift of every LayerNorm in the predictor (AdaLN-Zero [62], initialized to the identity). A projection of  $c_t$  is also appended to the cross-attention keys and values as one extra token. Four heads, one per supervision level, read the final query states.

### 3.6. Objective

With  $\rho$  the smooth- $\ell_1$  loss and LN a LayerNorm without affine parameters, the cross-modal loss for target  $t$  is

$$\mathcal{L}_{\text{cm}} = \frac{1}{L} \sum_{l=1}^L \frac{1}{G^2 D} \sum_{i,d} \rho(\hat{z}_{t,i,d}^l - \text{LN}(\text{sg}[\bar{z}_{t,i}^l])_d), \quad (4)$$

with  $L=4$  and sg the stop-gradient. Our latest recipe adds a dense context loss in the spirit of V-JEPA 2.1 [55]. Each visible source token at every level is pulled toward the EMA encoding of the full, unmasked source at the same cell. The weight of the token depends on its distance  $d_i$  (in cells) to the nearest hidden cell. V-JEPA 2.1 weights by  $1/\sqrt{d_i}$ ; we use an exponential that decays faster:

$$\mathcal{L}_{\text{ctx}} = \frac{1}{L} \sum_l \frac{1}{|V|} \sum_{i \in V} w_i \bar{\rho}(z_{s,i}^l, \text{LN}(\bar{z}_{s,i}^l)), \quad (5)$$

where  $\bar{\rho}$  averages  $\rho$  over features and  $w_i \propto e^{-(d_i-1)/\tau}$  with  $\tau=2$  cells, normalized to mean one. Tokens next to the hidden region, where the full view knows more than the masked one, carry most of the weight. The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{cm}} + \lambda_{\text{ctx}} \mathcal{L}_{\text{ctx}} + w_v \mathcal{V} + w_c \mathcal{C} + \alpha \mathcal{L}_{\text{moe}} + \beta \mathcal{L}_{\text{cond}}, \quad (6)$$

where  $\mathcal{V}$  and  $\mathcal{C}$  are the VICReg variance hinge and co-variance penalty [12]. Both are computed on the tokens of  $z^{\text{ctx}}$  and of the final-level prediction, flattened over the batch.  $\mathcal{L}_{\text{cond}}$  is an  $\ell_1$  loss for a small head that regresses the footprint-mean Diviner temperature (24 local-time bins) and GRAIL gravity from  $\lambda(\ell)$ . We use  $\lambda_{\text{ctx}}=0.5$ ,  $w_v=1$ ,  $w_c \in \{0.005, 0.01\}$ ,  $\alpha=10^{-3}$  and  $\beta=0.01$ . The EMA decay is 0.996, optionally ramped to 0.9995 with a cosine schedule. Section 4 explains why the regularizer weights are small, and why we came to distrust two of the terms.

## 4. When the loss lies

We trained a few dozen models with the objective of Sec. 3.6, and some failed in ways the loss did not show. This section describes the two failure modes (Fig. 2), explains why the usual health signals miss them and defines the monitor we now rely on.

### 4.1. Two failure modes

**Dimensional collapse.** In the run of Fig. 2 (fixed-GSD stems, seven products, peak learning rate  $10^{-4}$ ), the effective rank of the target tokens within a tile fell from 651 to 163 between 40k and 52k steps. The VICReg covariance rose from 3.7 to 13.6, the expert load drifted out of balance, and the median gradient norm went from 0.05 to over 200. The validation loss kept falling, from 0.280 to 0.149. This is the dimensional collapse known from joint-embedding methods [40, 45]: per-feature variance survives, but the representation concentrates in a few directions. Four of our runs failed this way (Tab. 2).

**Position-code collapse.** The second mode is harder to see. Between 5k and 35k steps of the run in Fig. 2, the validation loss fell from 0.389 to 0.038, the within-tile rank of targets and predictions rose from 382 to 667 and from 160 to 632, and the covariance penalty fell from 8.5 to 0.45. Yet the prediction for a footprint ended with cosine 0.972 to its own target and 0.971 to the target of another footprint. At every cell the model emitted a vector that depended on the cell’s position and hardly at all on the terrain. None of its low loss came from content, and every standard signal said the run had improved. Three runs ended this way (Tab. 2).

### 4.2. Why the usual signals are blind

Write a target token of product  $t$  at cell  $i$  of footprint  $x$  as

$$\bar{z}_t(i, x) = u(i) + m_t + l(\ell_x) + c_t(x, i), \quad (7)$$

a position term, a product term, a location term and the rest, which we call content. The predictor gets  $u$ ,  $m_t$  and  $l$  for free: rotary embeddings give it the query cell, the query carries the target product, and the location embedding is added to every token on both paths. Only  $c_t$  requires reading the source, and for many product pairs it is barely predictable at the token level (a 100 m image says little about kilometre-scale roughness). The student can lower the loss on those pairs only by shrinking  $c_t$  in its own features, and the EMA target follows. Predicting  $u+m_t+l$  and letting  $c_t$  fade is therefore a descent direction. In a same-image JEPA this pull is much weaker, because the context determines the masked target [3, 13].

This decomposition explains the blind spots in Tab. 2.

- *Loss* falls as  $c_t$  fades. Whatever cannot be predicted is removed from the target.

- *Within-tile rank and token cosine gap* are computed inside one tile, across positions. A position code has  $G^2$  distinct tokens per tile, so it scores near the maximum on both.
- *VICReg variance*, the per-feature standard deviation over the flattened tokens, is met by  $u(i)$  alone.
- *VICReg covariance* on flattened tokens is lowest for a position code. In a toy batch of 8 tiles of  $64^2$  layer-normalized tokens, a position code scores 0.19. Adding a shared content vector per tile raises it to 8.8, and tokens constant within each tile score 83. The shared per-tile component is what the penalty sees as correlated features. In our runs, the collapsed models ended at 0.44–0.45 and the healthy ones at 1.6–2.5.
- *Pooled SIGReg* [10], applied to the tile-mean prediction  $\bar{m}$ , has a scale conflict with Eq. (4). Layer-normalized tokens give  $\|\bar{m}\|^2 = D \bar{c}$ , where  $\bar{c}$  is the mean within-tile cosine. Matching  $\mathcal{N}(0, I)$  therefore needs  $\bar{c} \approx 1$ , that is, every token in a tile identical. In our runs with SIGReg (weight 0.003) the term sat at a constant  $\approx 67$  ( $\bar{c}=0.11$ ) and pushed only toward smoother tiles.

Rotary embeddings are relative, but our design still leaks absolute position. The illumination token sits at a fixed position outside the grid, so a query’s attention logit to it depends on the query’s absolute cell. At initialization a linear readout of the predictor output recovers the (row, col) of tokens in a held-out sample with  $R^2=0.86$  when this token is present (0.20 if the sun differs between samples) and  $R^2=-0.26$  without it. Windowed long-short attention [97] adds a second channel: its zero-padded border windows stamp the window class (corner, edge or interior) onto every token, 2.1 times the noise floor at initialization against 1.2 for standard attention. Both long-short runs in Tab. 2 collapsed as the learning rate approached its peak, the second despite a NATTEN-style border clamp [36] that cut the stamp to 1.06. The attractor is not specific to that architecture: a standard-attention model with a ten times larger covariance weight reached it within 4k steps.

### 4.3. A cross-sample monitor

A position code is invisible within a sample, so we compare across samples. For prediction  $\hat{z}_b$  and target  $\bar{z}_b$  of sample  $b$  in a micro-batch, and  $\pi(b)$  another sample of the same micro-batch, we log

$$\Delta = \mathbb{E}_b [\cos(\hat{z}_b, \bar{z}_b) - \cos(\hat{z}_b, \bar{z}_{\pi(b)})], \quad (8)$$

$$S = 1 - \Delta / \mathbb{E}_b [\cos(\hat{z}_b, \bar{z}_b)], \quad (9)$$

where  $\cos$  is averaged over the  $G^2$  cells and  $\bar{z}$  is layer-normalized.  $\Delta$  measures how much better a prediction matches its own footprint than another one, and  $S$  is the share of the agreement that another footprint’s target would have received anyway. A position code gives  $\Delta \rightarrow 0$  and

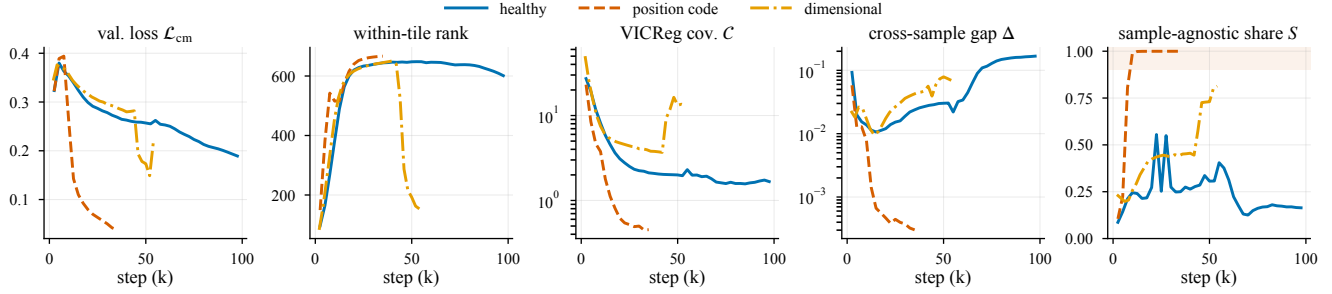


Figure 2. A healthy run, a position-code collapse and a dimensional collapse (validation curves of runs A1, B1 and D1 in Tab. 6, which use different recipes). During the position code the loss falls tenfold, the within-tile rank rises and the VICReg covariance drops, all apparent improvements, while the cross-sample gap  $\Delta$  goes to zero. In the dimensional collapse the rank falls and the covariance rises, but  $\Delta$  increases. Only  $S$  (Eq. (9)) rises in both; the band marks  $S > 0.9$ .

Table 2. Common signals while a run fails (runs B1, B2, C2, D1–D3 and E1 in Tab. 6), from the last healthy validation to the end of the failure. Rank: effective rank of the target tokens within one tile. Colors show what a practitioner would read into each change (looks better, looks worse). Only  $S$  moves the same, correct way in all seven.

| Run                                         | Mode        | Steps     | Val. loss $\mathcal{L}_{cm}$ | Within-tile rank | VICReg $C$ | Gap $\Delta$ | Share $S$ |
|---------------------------------------------|-------------|-----------|------------------------------|------------------|------------|--------------|-----------|
| Long-short attention [97]<br>+ border clamp | position    | 5k→35k    | .389→.038                    | 382→667          | 8.5→0.45   | .013→.0003   | .22→1.00  |
|                                             | position    | 5k→60k    | .393→.030                    | 369→665          | 8.0→0.44   | .017→.0003   | .36→1.00  |
|                                             | position    | 1k→8.5k   | .423→.429                    | 262→505          | 3.2→0.26   | .026→.003    | .23→.91   |
| Fixed-GSD, $S_{max}=600$                    | dimensional | 40k→52k   | .280→.149                    | 651→163          | 3.7→13.6   | .047→.074    | .46→.80   |
| Fixed-GSD, $S_{max}=1000$                   | dimensional | 58k→74k   | .262→.119                    | 625→90           | 3.2→19.2   | .025→.051    | .55→.89   |
| 50% NAC rows                                | dimensional | 25k→92.5k | .319→.089                    | 628→123          | 3.2→21.4   | .026→.024    | .29→.96   |
| Shared-canvas stems                         | dimensional | 27.5k→65k | .339→.164                    | 623→402          | 1.4→4.1    | .083→.127    | .05→.67   |

$S \rightarrow 1$ . Dimensional collapse makes all targets alike, so the cosine grows faster than the gap and  $S$  rises even while  $\Delta$  increases (Tab. 2). The cost is one extra cosine per target.

$S$  passed 0.65 in all seven failures and 0.8 in six of them. It is not a clean threshold. Runs that stayed useful ended between 0.16 and 0.57, but three of them passed 0.69–0.74 for a validation or two (Tab. 6). We read  $S < 0.35$  as healthy and  $S > 0.9$  as collapsed. In between, we watch whether  $\Delta$  is still rising. For position codes,  $S$  gave the earliest warning. It passed 0.7 at 7.5k steps, when the loss had not yet moved and the within-tile token cosine gap was still 0.06.

The monitor has limits.  $\Delta$  rewards anything that differs between footprints, including the location term  $l$  and, when  $\pi(b)$  holds a different product, the product term, so a code built from  $u$ ,  $m_t$  and  $l$  would pass. It cannot see predictions that stay specific to the footprint but lose detail within the tile. In one run the within-tile token cosine gap fell from 0.015 to 0.006 between 32.5k and 67.5k steps while  $\Delta$  held at 0.03–0.05, so we log both kinds of monitor.  $\Delta$  is identically zero at micro-batch size one, and it is measured against each run’s own EMA target, so absolute values compare only runs whose targets are alike. It is an alarm and a within-recipe score, not a substitute for downstream evaluation.

**A timing trap.**  $\Delta$  starts high (0.1–0.4 in the first 1–2k steps), because the targets of a fresh encoder are easy to tell apart, and then falls several-fold, with the variance hinge within a few hundred steps of the prediction standard deviation snapping to 1 (Sec. 5.1). Runs must be compared after this trough, at matched steps.

## 5. Experiments

**Setup.** The encoder has 217M parameters (104M active per token) and the predictor 116M. The stems add 1.7M (linear) to 62M (convolutional) over all 38 product slots, most of which a run never samples. We use AdamW [51] (weight decay 0.05, clipping at 1), bf16, linear warmup over 10% of the steps and cosine decay, on 16–64 A100 GPUs with 96 to 3072 footprint rows per step and flash attention [25] at  $G \geq 64$ . Every 2.5k steps we validate on a fixed, seeded subset of the validation footprints, with the block mask applied. Recipes changed as the project went on, so each comparison below changes one setting of one recipe (Appendix B). All results are single runs.

### 5.1. What pushes a run toward collapse

**The variance hinge fights calibration.** The VICReg variance term asks for a per-feature standard deviation of at least

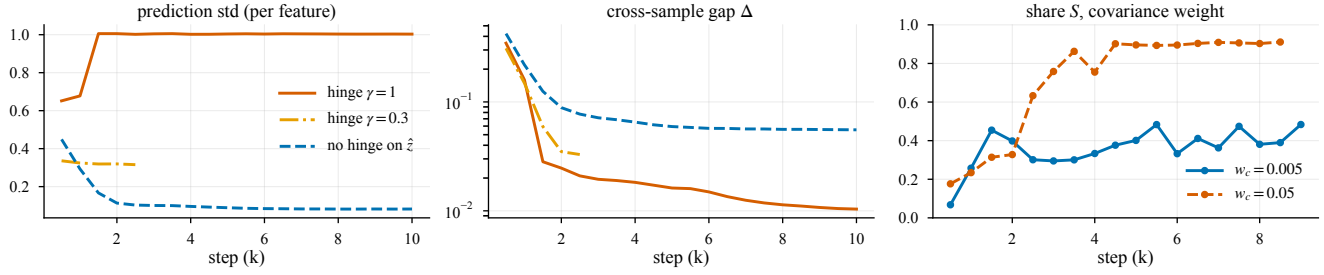


Figure 3. Two regularizers that pull the wrong way. *Left, middle*: with the variance hinge on the predictions ( $\gamma=1$ ), their per-feature standard deviation jumps to 1 at 1.5k steps and  $\Delta$  drops fivefold within 500 steps. Without it, the standard deviation settles near 0.08 and  $\Delta$  ends 5.3 times higher. The  $\gamma=0.3$  run stopped early. *Right*: a covariance weight of 0.05 sends a standard-attention model into a position code by 4k steps.

Table 3. One-variable comparisons:  $\Delta$  and  $S$  at the same step in both arms. Background recipes differ between rows; the runs are listed in Tab. 6. <sup>†</sup>The shared-canvas arm collapsed at this step and diverged shortly after. The fixed-GSD arm trained stably to 82.5k ( $\Delta=0.060$ ,  $S=0.38$ ).

| Change (baseline $\rightarrow$ variant)            | Step  | $\Delta$                | $S$                   |
|----------------------------------------------------|-------|-------------------------|-----------------------|
| Hinge on $\hat{z}$ : $\gamma=1 \rightarrow$ off    | 10k   | .010 $\rightarrow$ .055 | .77 $\rightarrow$ .24 |
| Hinge on $\hat{z}$ : $\gamma=1 \rightarrow 0.3$    | 2.5k  | .021 $\rightarrow$ .033 | .39 $\rightarrow$ .30 |
| Covariance $w_c$ : 0.005 $\rightarrow$ 0.05        | 8k    | .204 $\rightarrow$ .003 | .38 $\rightarrow$ .90 |
| 4 MoE layers $\rightarrow$ dense                   | 97.5k | .167 $\rightarrow$ .094 | .16 $\rightarrow$ .52 |
| Kernel floor $k_{\min}$ : 1 $\rightarrow$ 4        | 97.5k | .167 $\rightarrow$ .121 | .16 $\rightarrow$ .25 |
| Shared canvas $\rightarrow$ fixed-GSD <sup>†</sup> | 65k   | .127 $\rightarrow$ .045 | .67 $\rightarrow$ .56 |
| Grid $G$ : 64 $\rightarrow$ 32                     | 10k   | .010 $\rightarrow$ .009 | .77 $\rightarrow$ .55 |
| Grid $G$ : 64 $\rightarrow$ 128                    | 10k   | .010 $\rightarrow$ .016 | .77 $\rightarrow$ .63 |

$\gamma$  across tokens. That sounds harmless for layer-normalized targets, but a calibrated prediction  $\mathbb{E}[\hat{z} \mid \text{source}]$  has a standard deviation of about  $\sqrt{R^2}$ , where  $R^2$  is the share of the target’s variance the source explains. Ours is around 1%, so the hinge asks for ten times the calibrated spread. The cheapest variance to add without raising the loss is structure that the target also has and the predictor can produce without reading the source:  $u$ ,  $m_t$  and  $l$  of Eq. (7). With the hinge, the prediction standard deviation snaps to 1 and  $\Delta$  drops fivefold within 500 steps (Fig. 3). Turning it off for the predictions (it stays on the context) gives 5.3 times the gap at 10k steps,  $S$  of 0.24 instead of 0.77 and a lower validation loss (0.350 vs. 0.390), and  $\gamma=0.3$  falls in between. The hinge is on in every other run in this paper, including the downstream backbone, and probably explains part of their early trough.

**Covariance and experts.** Raising  $w_c$  from 0.005 to 0.05 turned a standard-attention model into a position code within 4k steps, as Sec. 4 predicts. Removing the four MoE layers lowered the late gap from 0.167 to 0.094 and raised  $S$  from 0.16 to 0.52, while the validation loss improved (0.168 vs. 0.189). The routers were not trouble-free, though: in

three long runs the first MoE layer lost balance before the gradient norm grew, and one run ended with three quarters of its experts unused.

## 5.2. Tokenization

**Fixed-GSD stems vs. a shared canvas.** With five products and nothing else changed, the shared-canvas model reached the higher gap (0.169 at 55k steps) but then collapsed dimensionally (rank 623  $\rightarrow$  402) and diverged at 66k. The fixed-GSD model held its rank near 610 and trained stably to the end of its allocation at 82.5k. The two stems produce different targets, so their gap values are not comparable; their stability is.

**Kernel floor.** The floor does what it was meant to do (Tab. 7). With  $k_{\min}=1$ , pairs involving LOLA roughness were close to chance ( $\Delta=0.033$  as target, 0.023 as source). With  $k_{\min}=4$  they rose three- to fourfold, and  $\text{TiO}_2$  gained as a target. The other pairs lost ground (0.205  $\rightarrow$  0.122), and so did the overall gap. With one seed per arm we cannot tell a trade-off from run-to-run variation. The fixed-GSD run we are now training follows the overall gap and uses  $k_{\min}=1$  [TODO: update when the  $G=64$  fixed-GSD run finishes].

**Grid size.** In a 10k-step sweep that changed only  $G$ , the validation loss improved with resolution (0.420, 0.390, 0.364 for  $G=32, 64, 128$ ), and  $G=128$  ended with the largest training-window gap on 11 of 12 target products. On 32 GPUs the runs took 24, 28 and 63 hours. All three had the variance hinge on and ended in the trough of Sec. 4.3, so the numbers rank the grids but understate them. We use  $G=64$  as the compromise.

## 5.3. Dense-prediction upgrades

The dense context loss (Eq. (5)) and the convolutional stems both aim at better features per cell. Trained together at  $G=32$  with shared-canvas stems and seven products, they raised  $\Delta$  from 0.19 at the first validation (2.5k steps) to 0.308

Table 4. Crater detection with a frozen encoder and fine-tuned stems (test split, single seed). *Top*: FCOS heads [84] with our neck. *Bottom*: the NASA-IBM LFM protocol (Faster R-CNN family, WAC only, boxes of at least 10 px, at most 500 detections).

| Data                                   | Input               | mAP <sub>50</sub> | mAP          |
|----------------------------------------|---------------------|-------------------|--------------|
| Robbins, 1k tiles                      | WAC                 | 0.530             | 0.238        |
|                                        | WAC + predicted DTM | 0.549             | 0.242        |
|                                        | WAC + DTM           | 0.595             | 0.285        |
| Robbins, 10k tiles                     | WAC                 | 0.530             | 0.270        |
|                                        | WAC + DTM           | <b>0.755</b>      | <b>0.411</b> |
| NAC, hand-labeled                      | NAC                 | 0.690             | 0.319        |
| NASA-IBM LFM [30], our reproduction    |                     | 0.464             | 0.190        |
| Ours, LFM detection stack              |                     | 0.362             | 0.156        |
| Ours, our neck + jitter, tokens only   |                     | 0.433             | 0.187        |
| Ours, our neck + jitter + pixel branch |                     | <b>0.494</b>      | <b>0.217</b> |

at 10k instead of dropping into the trough, and held 0.294 at 35k, with  $S$  between 0.05 and 0.23. The closest earlier run shares grid, products, learning rate, batch and regularizers and reached  $\Delta=0.064$  ( $S=0.45$ ) at 35k. It also differs in normalization statistics, validation subset and code version, so the comparison is confounded. Clean ablations and the  $G=64$  runs on both stem types are in progress [TODO: fill in the two  $G=64$  runs and a control with both upgrades off].

## 5.4. Downstream

The downstream results use an earlier checkpoint ( $G=64$ , shared-canvas stems, six products, 79k steps at a peak learning rate of  $10^{-4}$ , variance hinge on). Its transformer is always frozen. The crater detectors fine-tune the per-product stems with their heads, and the probes read frozen features only (Appendix D).

**Crater detection.** We train detection heads for the Robbins catalogue [70] on WAC tiles at 100m (1k and 10k tiles) and for hand-labeled NAC craters at  $\approx 1$  m (Tab. 4). Adding the DTM as a second input to the same frozen backbone is the largest single effect:  $+0.065$  mAP<sub>50</sub> with 1k tiles and  $+0.225$  with 10k tiles, where the lighting varies more. A DTM that the backbone itself predicts from the WAC image recovers 0.019 of the 0.065. For a comparison with the NASA-IBM lunar foundation model [30], which is trained by masked token prediction, we reproduced its frozen-backbone Faster R-CNN protocol on the same data. We obtained 0.464 mAP<sub>50</sub> for that model, against 0.458 reported. With its detection stack our tokens reach 0.362–0.390. With our neck and scale jitter they reach 0.433 from tokens alone, and 0.494 when the detector also gets a raw-pixel branch. Our fine-tuned stems favor us in this comparison, so we read it as rough parity on single-product craters. The distinctive gain comes from the second product.

**Shape, albedo and shadows.** A WAC image mixes topography, albedo and illumination, and the backbone was never trained to separate them. We probe whether its predictions do, on 40 held-out footprints seen under eight suns each (full metrics in Tab. 8). Given only a WAC image, linear probes on the predictor features for the elevation and 643 nm products recover relief with correlation 0.705 (0.12 for the same probe on pixels) and albedo with 0.840. Both barely change with the sun, and the predicted relief picks the right footprint among the 40 in 96% of cases. A small head that integrates the backbone’s own slope and aspect predictions reaches a relief RMSE of 299 m (551 m for zero relief). The reverse direction tests the sun conditioning. From topography and a sun, a linear probe on the predictor’s WAC features localizes cast shadows with AUC 0.89, against 0.59 with another observation’s sun and 0.49 with none. The predictor uses the sun it is given.

## 6. Limitations and conclusion

**Limitations.** All comparisons are single runs, and several change one setting on top of recipes that differ from each other. The downstream backbone predates the fixed-GSD stems, the dense context loss and the convolutional stems, so the current recipe is not yet evaluated downstream [TODO: final checkpoint]. The cross-sample gap cannot separate content from location or product identity, and it compares only runs whose targets are alike. The mechanism of Eq. (7) is supported by probes and ablations but not proven. Polar regions and the 1 m NAC scale, where much of the scientific demand lies, are outside the runs reported here.

**Conclusion.** Predicting one lunar instrument from another in latent space is a workable pretraining task. Its frozen features detect craters about as well as a masked-token lunar model does, gain substantially from a second input product and respond to the sun they are given. Getting there taught us more about failure than about architecture. A cross-modal JEPa can reach a very low loss by encoding where a token is instead of what the ground looks like, and loss, within-sample rank and VICReg all read this as progress. A cross-sample comparison exposes it for one cosine per target, and we would log it in any JEPa whose targets are only partly predictable from the context.

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# LeLunar: Learning the Moon by Predicting One Instrument from Another

## Supplementary Material

### A. Receptive-field offset of strided conv stems

Consider one stage with stride  $s$ , kernel  $k$  and padding  $q$  on a one-dimensional input of length  $sn$  (pixels  $0, \dots, sn-1$ ). Output  $o$  reads input pixels  $so-q, \dots, so-q+k-1$ , so its receptive field is centered at  $so - q + (k-1)/2$ . The cell that output  $o$  stands for spans pixels  $so, \dots, so+s-1$ , with center  $so + (s-1)/2$ . The offset of one stage is therefore

$$\delta_1 = \frac{k-1}{2} - q - \frac{s-1}{2}. \quad (10)$$

With  $k=3$  and  $q=1$ ,  $\delta_1 = -(s-1)/2$  input pixels. Stacking stages with strides  $s_1, \dots, s_n$  and total stride  $P = \prod_j s_j$  gives, in native pixels,

$$\delta = -\sum_j \left( \prod_{i < j} s_i \right) \frac{s_j - 1}{2} = -\frac{P-1}{2}, \quad (11)$$

because the sum telescopes. In units of one token cell ( $P$  pixels) this is Eq. (2). With  $k=s+2$  and  $q=1$ ,  $\delta_1=0$  for every  $s$ , the receptive field extends one pixel beyond the cell on each side, and the output length is unchanged. At  $s=3$  the  $3 \times 3$  kernel reads pixels  $3o-1, \dots, 3o+1$ . Pixel  $3n-1$  is never read, and two such stages leave the last four of 576 columns of a WAC tile unused. Stride-2 stages read every pixel at either kernel size.

Table 5. Measured token offset (fraction of a cell, negative is up-left) and unread border columns with kernel 3, for the conv stems at  $G=64$ .

| $p$ (strides)   | kernel 3 | kernel $s+2$ | unread cols |
|-----------------|----------|--------------|-------------|
| 4 (2, 2)        | -0.375   | 0            | 0           |
| 9 (3, 3)        | -0.444   | 0            | 4           |
| 16 (2, 2, 2, 2) | -0.469   | 0            | 0           |

### B. Runs

Table 6 lists every run this paper cites. All runs use the ViT-B encoder with four MoE layers (except A3) and the 12-layer predictor, four supervision levels,  $K=2$  targets and a visible block of 70–80% (except B1, B2, which see the full source). The variance hinge is on for context and predictions with  $\gamma=1$  except in F4 and F5. SIGReg is off in every run listed.  $\Delta$  and  $S$  are validation values at the last step; max  $S$  is taken after 5k steps.

**Shared recipe.** Unless stated otherwise: ViT-B encoder with the last four blocks MoE (8 experts, top-2), 12-layer

predictor of width 768, four supervision levels, visible square block covering 70–80% of the grid,  $K=2$  targets per row, VICReg variance weight 1 on context and predictions, covariance 0.005–0.01, no SIGReg, EMA decay 0.996 (constant or ramped to 0.9995), AdamW with weight decay 0.05 and clipping at 1, warmup over 10% of the schedule, cosine decay to  $10^{-6}$ .

**Runs in Tab. 2.** The two long-short attention runs use  $G=64$ , fixed-GSD stems with  $k_{\min}=1$ , 12 products, no source masking, peak learning rate  $3 \times 10^{-5}$  and  $w_c=0.01$ . The second adds the border clamp and uses  $S_{\max}=4096$  instead of 600. The  $w_c=0.05$  run uses  $G=32$ , shared-canvas stems, 5 products and peak learning rate  $10^{-4}$  at an effective batch of 3072, and differs from its  $w_c=0.005$  twin only in that weight. The twin is not a clean healthy reference: over its last 4k steps, before it stopped at 9.3k, its within-tile rank fell from 384 to 120 while its covariance rose from 5.7 to 14.8, the early signature of dimensional collapse, with  $S$  at 0.38. The pair shows only that the larger weight produced a position code. The fixed-GSD pair uses  $G=64$ , seven products,  $k_{\min}=1$  and peak learning rate  $10^{-4}$ , and differs only in  $S_{\max}$ . The 50%-NAC run uses 12 products at  $G=64$  and draws half of its rows from Level-2 NAC subtiles. The shared-canvas run is the baseline arm of the five-product stem comparison of Sec. 5.2.

**Downstream backbone.**  $G=64$ , shared-canvas stems, products WAC visible, WAC UV, elevation, slope, aspect and the 643 nm mosaic. LOLA roughness was configured but dropped by the raster-size gate of the shared-canvas path.  $S_{\max}=600$ , peak learning rate  $10^{-4}$ , constant EMA 0.996,  $w_c=0.005$ , effective batch 512, 79.1k steps, final  $\Delta=0.046$  and  $S=0.57$ .

### C. Kernel floor per product pair

Table 7 breaks the kernel-floor comparison of Sec. 5.2 (runs A1 and A2) down by groups of product pairs.

### D. Downstream protocols

Table 8 lists all relief, albedo and shadow probe metrics of Sec. 5.4.

**Crater detection.** Robbins 1k has 800/100/100 WAC tiles of 512 px at 100 m, Robbins 10k has 8000/1000/1000 tiles with a wider range of illumination, and the NAC set has 289/63/56 hand-labeled tiles of 256 px at about 1 m. The

Table 6. Runs cited in the paper. Stems: shared canvas (C) or fixed-GSD with kernel floor  $k_{\min}$  ( $Fk$ ). EMA: constant 0.996 (c) or cosine ramp to 0.9995 (r). Batch: effective footprint rows per optimizer step. <sup>‡</sup>Seven products configured; the raster gate of the shared-canvas path drops LOLA roughness at  $G=64$ , leaving six.

| ID | Run                                 | $G$ | Prod.          | Stems | $S_{\max}$ | LR   | Batch | EMA | $w_c$ | Steps  | $\Delta$ | $S$  | max $S$ / outcome           |
|----|-------------------------------------|-----|----------------|-------|------------|------|-------|-----|-------|--------|----------|------|-----------------------------|
| A1 | MoE, $k_{\min}=1$                   | 64  | 12             | F1    | 1000       | 3e-5 | 256   | r   | 0.01  | 97.5k  | .167     | .16  | .56 / healthy               |
| A2 | MoE, $k_{\min}=4$                   | 64  | 12             | F4    | 1000       | 3e-5 | 256   | r   | 0.01  | 97.5k  | .121     | .25  | .42 / healthy               |
| A3 | dense, $k_{\min}=1$                 | 64  | 12             | F1    | 1000       | 3e-5 | 256   | r   | 0.01  | 112.5k | .119     | .51  | .71 / healthy               |
| B1 | long-short attention                | 64  | 12             | F1    | 600        | 3e-5 | 256   | r   | 0.01  | 35k    | .000     | 1.00 | 1.00 / position code        |
| B2 | long-short + border clamp           | 64  | 12             | F1    | 4096       | 3e-5 | 256   | r   | 0.01  | 60k    | .000     | 1.00 | 1.00 / position code        |
| C1 | $w_c=0.005$                         | 32  | 5              | C     | 600        | 1e-4 | 3072  | c   | 0.005 | 9k     | .181     | .48  | .48 / rank falling, stopped |
| C2 | $w_c=0.05$                          | 32  | 5              | C     | 600        | 1e-4 | 3072  | c   | 0.05  | 8.5k   | .003     | .91  | .91 / position code         |
| D1 | $S_{\max}=600$                      | 64  | 7              | F1    | 600        | 1e-4 | 96    | c   | 0.005 | 54k    | .069     | .81  | .81 / dimensional           |
| D2 | $S_{\max}=1000$                     | 64  | 7              | F1    | 1000       | 1e-4 | 96    | c   | 0.005 | 118k   | .120     | .66  | .90 / dimensional           |
| D3 | 50% NAC rows                        | 64  | 12             | F1    | 1000       | 1e-4 | 256   | c   | 0.01  | 160k   | .024     | .97  | .99 / dimensional           |
| E1 | shared canvas, 5 prod.              | 64  | 5              | C     | 1000       | 5e-5 | 128   | c   | 0.01  | 65k    | .127     | .67  | .67 / dimensional, diverged |
| E2 | fixed-GSD, 5 prod.                  | 64  | 5              | F1    | 1000       | 5e-5 | 128   | c   | 0.01  | 82.5k  | .060     | .38  | .69 / healthy               |
| F1 | grid sweep                          | 32  | 12             | F4    | 4096       | 3e-5 | 256   | r   | 0.01  | 10k    | .009     | .55  | .55 / short                 |
| F2 | grid sweep                          | 64  | 12             | F4    | 4096       | 3e-5 | 256   | r   | 0.01  | 10k    | .010     | .77  | .77 / short                 |
| F3 | grid sweep                          | 128 | 12             | F4    | 4096       | 3e-5 | 256   | r   | 0.01  | 10k    | .016     | .63  | .63 / short                 |
| F4 | F2, no hinge on $\hat{z}$           | 64  | 12             | F4    | 4096       | 3e-5 | 256   | r   | 0.01  | 10k    | .055     | .24  | .24 / short                 |
| F5 | F2, hinge $\gamma=0.3$ on $\hat{z}$ | 64  | 12             | F4    | 4096       | 3e-5 | 256   | r   | 0.01  | 2.5k   | .033     | .30  | – / stopped                 |
| G1 | dense loss + conv stems             | 32  | 7              | C     | 600        | 1e-4 | 768   | c   | 0.005 | 40k    | .283     | .23  | .23 / healthy, running      |
| G2 | closest earlier run                 | 32  | 7              | C     | 600        | 1e-4 | 768   | c   | 0.005 | 48.5k  | .062     | .51  | .67 / healthy               |
| H  | downstream backbone                 | 64  | 6 <sup>‡</sup> | C     | 600        | 1e-4 | 512   | c   | 0.005 | 79k    | .046     | .57  | .74 / used downstream       |
| I  | within-tile smoothing               | 64  | 12             | F1    | 1000       | 5e-5 | 128   | c   | 0.01  | 87.5k  | .082     | .55  | .67 / smoothing             |

Table 7. Kernel floor, per group of product pairs (validation  $\Delta$  at 97.5k steps, macro average over pairs; 12 products, grid 64). The floor affects LOLA roughness (1 px/token at  $k_{\min}=1$ ) and the 500 m and 400 m WAC products (2 px/token).

| Pairs                          | #pairs | $k_{\min}=1$ | $k_{\min}=4$ |
|--------------------------------|--------|--------------|--------------|
| Roughness as target            | 11     | 0.033        | <b>0.103</b> |
| Roughness as source            | 11     | 0.023        | <b>0.100</b> |
| TiO <sub>2</sub> as target     | 11     | 0.079        | <b>0.112</b> |
| WAC UV as target               | 11     | <b>0.145</b> | 0.134        |
| Pairs without floored products | 42     | <b>0.205</b> | 0.122        |
| Overall (count-weighted)       | 132    | <b>0.167</b> | 0.121        |

Table 8. Relief, albedo and shadows from frozen features (40 held-out footprints, 8 suns each). Invariance: correlation between predictions for one footprint under different suns. Retrieval: top-1 accuracy of matching a prediction to its footprint.

| Probe and metric                             | Value         |
|----------------------------------------------|---------------|
| <i>WAC → relief, linear probe</i>            |               |
| correlation (same probe on pixels: 0.12)     | 0.705         |
| sun invariance / retrieval                   | 0.90 / 0.96   |
| <i>WAC → albedo, linear probe</i>            |               |
| correlation with the 643 nm mosaic           | 0.840         |
| sun invariance / retrieval                   | 0.87 / 1.00   |
| <i>WAC → relief, SAFS head</i>               |               |
| correlation / RMSE (zero relief: 551 m)      | 0.808 / 299 m |
| <i>DTM + sun → WAC shadows, linear probe</i> |               |
| AUC with the true sun                        | <b>0.89</b>   |
| AUC with another observation’s sun           | 0.59          |
| AUC with no sun                              | 0.49          |

mid. The per-product stems are fine-tuned with the neck and head at learning rate  $10^{-4}$ , while the transformer stays frozen. We train with multi-scale native crops. The pixel branch is a shallow convolution on the raw input that feeds the finest pyramid level. Our protocol scores all boxes down to 2 px with at most 300 detections. The NASA-IBM protocol keeps boxes with a shortest side of at least 10 px, drops tiles without craters and allows 500 detections. In that protocol the NASA-IBM backbone is fully frozen, whereas our stems are fine-tuned, a difference in our favor. The predicted DTM comes from a DPT head trained on the backbone’s elevation-product predictor features, given the WAC image.

**Relief, albedo and shadows.** Footprints with at least six suns are split, with a fixed seed, into 120 for fitting and 40 for evaluation. Outputs are 128 px (469 m per pixel). The relief target is the plane-detrended 60 m DTM. Albedo is scored against the 643 nm mosaic and, separately, against a multi-sun pseudo-albedo. The linear probes read frozen predictor features. The relief head fuses the elevation-product features across depths, is conditioned on the sun, and integrates predicted slope and aspect with a Frankot–Chellappa solver. It was fitted on 600 footprints with two suns each. Shadow AUC uses the darkest-quantile pixels of the WAC image as labels, on high-incidence samples only, where dark pixels are mostly cast shadow.

detectors read the frozen EMA encoder at its four supervision depths through a trainable three-level feature pyra-